Serum gp100 as an Independent Prognostic Biomarker in Early-Stage Uveal Melanoma

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Purpose. To evaluate the prognostic significance of serum gp100 levels in patients with uveal melanoma (UM) without detectable metastases at diagnosis.

Methods. This prospective study included 37 patients with UM and no clinical evidence of metastasis at baseline. Serum gp100 concentration was quantified using a commercial ELISA kit. Tumors were molecularly profiled and categorized into high- or low-risk classes. Patients were stratified based on the median gp100 concentration (1.23 ng/mL). Survival analysis was performed using Kaplan–Meier estimates, and multivariate logistic regression was used to identify independent predictors of metastasis.

Results. During a mean follow-up of 24.6 months, 14 patients (37.8%) developed metastases. Patients with baseline gp100 >1.23 ng/mL had significantly shorter metastasis-free survival ($P < 0.001$). Receiver operating characteristic (ROC) analysis yielded an area under the curve of 0.89, with a cutoff of 1.387 ng/mL, achieving 85.7% sensitivity and 82.6% specificity. In multivariate analysis, serum gp100 remained an independent predictor of metastasis (odds ratio = 3849.9; 95% confidence interval, 9.66–1,534,206; $P = 0.007$), regardless of molecular classification. No significant correlations were found between gp100 and clinical or genetic parameters.

Conclusions. Elevated serum gp100 is an independent biomarker of early metastatic risk in UM, even in the absence of clinical or genetic high-risk features. ELISA-based measurement of gp100 may support noninvasive stratification and personalized surveillance strategies in clinical practice.

Keywords: gp100, biomarker, uveal melanoma

Uveal melanoma (UM) is the most common primary intraocular malignancy in adults and accounts for 85% to 90% of ocular melanomas. Although rare, with an annual incidence of five to seven cases per million in Europe, UM carries a disproportionate burden owing to its aggressive clinical course and high rate of metastasis.[1,2] Despite the frequently successful local control of the primary tumor through brachytherapy or enucleation, up to 50% of patients develop systemic dissemination, most commonly to the liver.[2] With the establishment of metastases, the prognosis remains poor, and median overall survival is 6 to 12 months.[3–5] A major challenge in the management of UM is the early identification of patients who are at risk of developing metastatic disease. Current prognostic stratification relies on chromosomal abnormalities (e.g., monosomy 3, 8q gain), mutational status (e.g., *BAP1*, *SF3B1*, *EIF1AX*), and gene expression profiling, which classify tumors into high- and low-risk categories.[6–9] Although these tumor-based analyses provide valuable insights, they are invasive, require tumor tissue that is frequently obtained by fine-needle aspiration biopsy, and offer only a static snapshot of tumor biology. Moreover, they may not capture the functional or immunological behavior of the tumor, which evolves over time.

In recent years, liquid biopsy tools such as circulating tumor DNA (ctDNA) and circulating tumor cells (CTCs) have emerged as promising noninvasive methods for monitoring tumor burden and disease progression. However, their sensitivity in early-stage or low-volume disease remains suboptimal, and their technical requirements limit their widespread clinical utilization.[10–12] Recent advances in proteomic analysis have opened new avenues for the identification of prognostic biomarkers in uveal melanoma. Liquid biopsy sources such as blood, aqueous humor, and vitreous fluid offer a minimally invasive means of assessing tumor biology in real time. These samples are amenable to serial collection, enabling dynamic monitoring of metastatic risk and treatment response. The detection and validation of circulating protein signatures, including gp100, across different platforms reinforce their potential clinical utility for early screening, prognosis, and therapeutic guidance.[13–16]

gp100, also known as PMEL or ME20, is a transmembrane glycoprotein selectively expressed in melanocytes and melanoma cells that plays a key structural role in melanosome maturation by promoting the formation of intraluminal fibrillar scaffolds that are required for melanogenesis. The luminal domain of gp100 undergoes proteolytic processing to generate amyloid-like fibrils that serve as a matrix for melanin deposition.[17,18] gp100 is transcriptionally regulated by MITF and is considered a hallmark melanocytic differentiation antigen. Importantly, its intracellular processing induces the presentation of peptide fragments by HLA class I molecules, which makes it a highly immunogenic, clinically validated target in cancer immunotherapy.[19]

This functional relevance has been translated into the development of tebentafusp, a bispecific T-cell receptor (TCR) fusion protein that redirects $CD3^+$ T cells toward gp100–HLA-A*02:01 complexes on melanoma cells. Tebentafusp is the first and, to date, only systemic therapy to demonstrate a significant overall survival benefit in metastatic UM, underscoring the clinical importance of gp100 as a stable, targetable tumor antigen.[20–22] Despite this therapeutic validation, the potential role of gp100 as a circulating biomarker of prognosis has been scarcely explored. Its detectability in serum and association with tumor activity, particularly in early disease, remain poorly understood. Prior work by our group has shown that serum gp100 levels are elevated in patients with UM compared to controls and even more markedly in those with metastatic disease.[23] However, whether serum gp100 can serve as an independent predictor of metastatic progression in nonmetastatic patients at diagnosis has not been previously addressed.

This study aimed to evaluate the prognostic value of serum gp100 levels in patients with UM without detectable metastasis at diagnosis. We hypothesized that elevated serum concentrations of gp100 would be associated with an increased risk of developing metastases and may provide complementary prognostic information to traditional clinical and genetic markers.

Methods

Study Design and Patients

This was a prospective, observational, prognostic study including 37 consecutive patients diagnosed with UM at the Ocular Oncology Unit of the Complejo Hospitalario Universitario de Santiago, Spain, between January 2016 and December 2018. All patients were free of clinical metastases at the time of inclusion. Blood samples were collected at the time of local treatment, either with brachytherapy or enucleation.

The inclusion criteria were a confirmed diagnosis of UM, no clinical evidence of metastasis at baseline, and availability of tumor and serum samples. Exclusion criteria included prior malignancies, active systemic disease, cutaneous melanocytic lesions, or incomplete follow-up. A minimum follow-up period of 12 months was required.

No patients presented with extraocular extension, either clinically or on histopathology. One tumor showed limited iris involvement originating from the ciliary body. All local treatments (brachytherapy, enucleation, endoresection, or proton therapy) achieved successful local tumor control, with no evidence of local recurrence during follow-up.

The study was approved by the Galician Clinical Research Ethics Committee (Code 2009/128) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants.

Clinical and Ophthalmologic Evaluation

Baseline ophthalmologic evaluation included best-corrected visual acuity (Snellen chart), slit-lamp biomicroscopy, indirect ophthalmoscopy, and multimodal imaging (Topcon TRC 50EX; Clarus, Carl Zeiss Meditec). Tumor height and basal diameter were measured by B-scan ultrasonography (Quantel Absolu, 15–20 MHz). Macular involvement was assessed using spectral-domain OCT (Heidelberg Engineering, Heidelberg, Germany). Systemic staging included liver ultrasound, chest X-ray, and liver function tests, repeated every 6 months.

Tumor location (choroid, ciliary body, peripapillary, or posterior pole) was recorded based on clinical and imaging features. One patient presented with a tumor involving the ciliary body and extending into the iris; this case was included in the ciliary body group for statistical purposes. Pigmentation (melanotic, amelanotic, or mixed) and the presence of exudative retinal detachment were also recorded. Tumors were classified according to the T category of the AJCC eighth edition[24] based on ultrasonographic measurements. Tumor volume was estimated using the formula for a prolate ellipsoid: (π × height × [basal diameter]^2) / 6, as used in prior UM volumetric studies.[25,26]

The primary endpoint was the occurrence of metastatic disease during follow-up, confirmed by imaging or histopathologic diagnosis.

Serum Collection and gp100 Quantification

Peripheral venous blood was collected in SST II Advance Vacutainer tubes (Becton-Dickinson, USA), allowed to clot for 30 minutes, and centrifuged at 1500 × g for 15 min. The serum was aliquoted and stored at –80°C until analysis.

Serum gp100 levels were quantified using a commercial ELISA kit (A2341; Antibodies.com, Stockholm, Sweden) according to the manufacturer's protocol. Samples were diluted 1:2 and analyzed in duplicate. Standard curves and internal quality controls were included on each plate. The assay range was 0.156 to 10 ng/mL, with a detection limit of 0.061 ng/mL. Results were excluded if intra-assay variation exceeded 10%.

Serum gp100 concentrations were analyzed both as a continuous variable and as a categorical variable using the cohort median (1.23 ng/mL) as the cutoff. This data-driven threshold was selected prospectively to avoid outcome-based stratification.

Genetic and Molecular Analysis

Tumor samples were obtained at the time of treatment via transscleral or endoresection biopsy. MLPA (SALSA MLPA kit; MRC-Holland) was used to assess chromosomal alterations. Mutational analysis was performed for *GNAQ*, *GNA11*, *BAP1*, *SF3B1*, and *EIF1AX*. Based on molecular profile, tumors were classified into low-risk or high-risk categories. Tumors were classified as high risk if they presented with monosomy 3, based on established molecular prognostic criteria. All other tumors were considered low risk.

Statistical Analysis

Statistical analyses were conducted using SPSS version 27 (IBM Corp., Chicago, IL, USA). The distribution normality was assessed using the Shapiro–Wilk test. Continuous variables were compared using Student's *t*-test or the Mann–Whitney *U* test, and categorical variables were compared using χ^2 or Fisher's exact tests, as appropriate. The correlation between tumor volume and gp100 levels was assessed using Spearman's rank correlation.

Kaplan–Meier survival analysis with log-rank testing was used to compare metastasis-free survival according to gp100 groups (above versus below median). Neither multivariate models nor receiver operating characteristic (ROC) curve analyses were performed due to the limited number of metastatic events and the exploratory nature of this study. A two-sided $P < 0.05$ was considered statistically significant.

Results

Patient and Tumor Characteristics

The mean age of the cohort was 69.0 ± 12.7 (range, 41–91) years, with a predominance of male patients (64.9%). Tumors were most commonly located in the peripheral choroid (48.6%). The average tumor height was 7.1 ± 3.8 mm, and the mean basal diameter was 12.9 ± 3.3 mm. A strong positive correlation was found between these two dimensions (Spearman's $r = 0.636$, $P < 0.001$), which indicated that larger tumors tended to have both greater height and basal extension. Regarding morphologic features, most lesions exhibited a nodular configuration (70.3%) and medium echogenicity on B-scan ultrasonography (75.7%).

Treatment and Genetic Profile

Initial treatment modalities included enucleation in 48.6% of cases, iodine-125 brachytherapy in 37.8%, endoresection in 10.8%, and proton beam therapy in 2.7%. Tumor samples for molecular analysis were obtained during the primary intervention.

In this cohort, chromosomal alterations were common, with monosomy 3 and 8q gain observed in 50% and 47.1% of patients, respectively. Mutational analysis identified pathogenic variants in *GNAQ* (51.4%), *GNA11* (35.1%), *BAP1* (29.7%), *SF3B1* (18.9%), and *EIF1AX* (16.2%). Overall, 50% of tumors were classified as high risk.

Follow-up and Metastatic Progression

After a mean follow-up of 24.6 ± 6.8 months, 14 patients (37.8%) developed metastatic disease, with the liver being the initial site of tumor dissemination in 92% of cases. The time from diagnosis to detection of metastasis was 9 to 41 months. Following progression, treatments varied and included palliative care (45.4%), systemic immunotherapy (18.2%), combination immunotherapy and chemotherapy (18.2%), hepatic surgery (9.1%), and treatment refusal (one patient, 4.5%). The analysis of potential prognostic variables revealed no significant intergroup differences in baseline clinical features between patients with and without metastasis. No significant differences were observed between groups in terms of age ($P = 0.266$), sex ($P = 0.724$), tumor height ($P = 0.938$), or basal diameter ($P = 0.863$). Conversely, several genetic markers showed notable associations with metastatic progression. Monosomy 3, which corresponds to the high-risk molecular class, was significantly more frequent in patients who developed metastases compared to those who did not (78.6% vs. 26.1%, $P = 0.017$). Similarly, 8q gain was more prevalent in the metastatic group (64.3% vs. 30.4%), although the difference did not reach statistical significance ($P = 0.097$). BAP1 mutations were also slightly more common in the metastatic group (42.9% vs. 21.7%), but this difference was not statistically significant ($P = 0.173$).

Serum gp100 Levels

The mean serum gp100 level was 1.32 ± 0.55 ng/mL (range, 0.45–3.41), with a nonnormal distribution (Shapiro–Wilk $P < 0.05$). The median was 1.23 ng/mL (interquartile range [IQR], 0.95–1.67), which was used as a non-outcome-based threshold to stratify the cohort (Table).

Patients who developed metastases had significantly higher baseline gp100 levels than those who remained disease-free (median, 1.76 [IQR, 1.72–2.23] ng/mL vs. 1.05 [IQR, 0.88–1.43] ng/mL, $P < 0.001$). A nonsignificant negative correlation was observed between gp100 concentration and time to metastasis (Spearman's $\rho = -0.372$, $P = 0.172$), suggesting a trend toward earlier progression in patients with elevated gp100 levels (Fig. 1).

No significant associations were observed between serum gp100 and other baseline clinical variables, including age, sex, tumor location, height, basal diameter, or AJCC T category ($P = 0.743$). Similarly, gp100 levels did not significantly

Table. Baseline Clinical, Anatomic, and Genetic Features of Patients Stratified by Serum gp100 Concentration (Cutoff = 1.23 ng/mL, Cohort Median)

Variable	gp100 ≤ 1.23 (*n* = 19)	gp100 > 1.23 (*n* = 18)	*P* Value
Age, y	66 ± 13	73 ± 12	0.110
Male sex	13 (68.4)	11 (61.1)	0.737
Tumor location			
Posterior/peripapillary	13 (68.4)	11 (61.1)	0.295
Choroid	1 (5.3)	4 (22.2)	
Ciliary body	5 (26.3)	3 (16.7)	
T category			
T1–T2	7 (36.8)	8 (44.4)	0.743
T3–T4	12 (63.2)	10 (55.6)	
Tumor pigmentation			
Amelanotic	2 (10.5)	3 (16.7)	0.547
Melanotic	16 (84.2)	15 (83.3)	
Mixed	1 (5.3)	0 (0.0)	
Exudative detachment			
No	13 (68.4)	13 (72.2)	1.000
Yes	6 (31.6)	5 (27.8)	
Tumor height, mm	7.2 ± 3.5	7.03 ± 4.3	0.775
Basal diameter, mm	13.42 ± 3.1	12.40 ± 3.6	0.461
Tumor volume, mm^3	797.5 ± 711	737.5 ± 629	0.641
Monosomy 3/high-risk molecular class	6 (35.3)	11 (64.7)	0.169
8q gain	5 (29.4)	11 (64.7)	0.084
BAP1 mutation	6 (31.6)	5 (27.8)	1.000
Metastasis	2 (10.5)	12 (66.7)	**<0.001**

Continuous variables are shown as mean ± SD, and categorical variables are shown as counts and percentages. Statistical comparisons used Mann–Whitney *U* and Fisher's exact tests. Metastatic progression was significantly more frequent in the high gp100 group ($P < 0.001$).

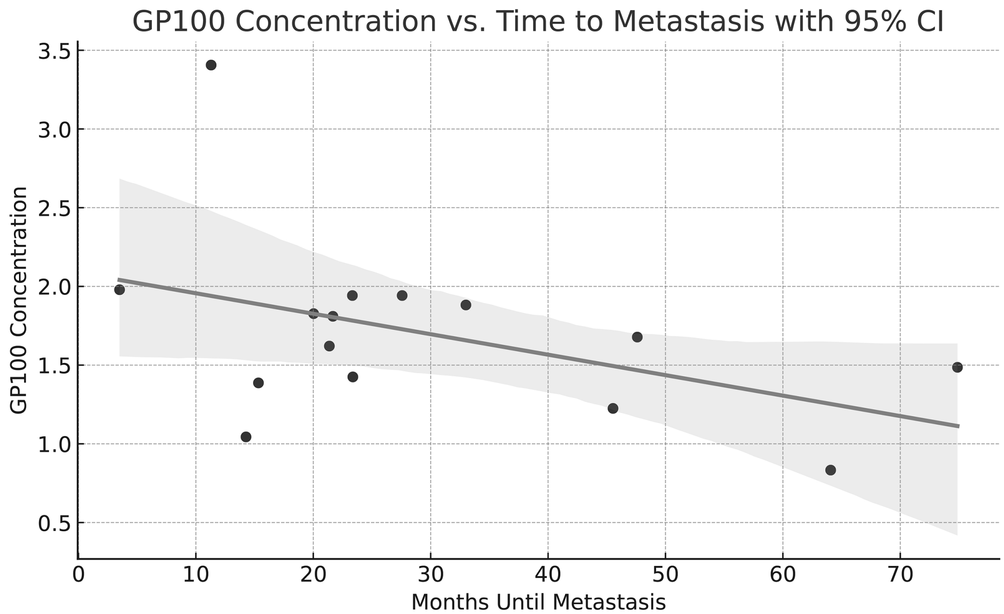

Figure 1. The relationship between the gp100 concentration and time (in months) until the onset of metastasis. A decreasing trend is observed, which suggests that higher gp100 levels may be associated with an earlier onset of metastasis. The regression line includes the 95% confidence interval.

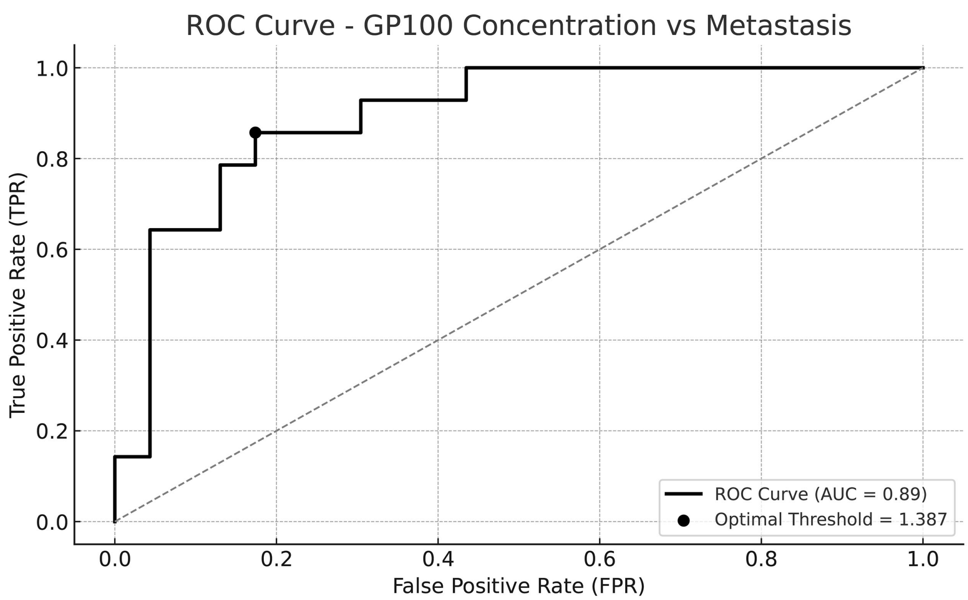

Figure 2. The ROC curve for the assessment of the discriminatory ability of the gp100 concentration in patients with and without metastasis. The AUC of 0.89 indicates good discriminative performance. The optimal threshold was 1.387 and yielded a sensitivity of 85.7% and a specificity of 82.6%.

differ by tumor pigmentation ($P = 0.547$) or the presence of exudative retinal detachment ($P = 1.000$). The correlation between gp100 and tumor volume was weak and not statistically significant ($r = 0.159$, $P = 0.345$; Table).

No significant differences in gp100 levels were observed based on molecular alterations, including monosomy 3 ($P = 0.169$), 8q gain ($P = 0.084$), or *BAP1* mutation ($P = 1.000$), as detailed in the Table. In the exploratory analysis, ROC curve evaluation of serum gp100 for predicting metastatic development yielded an area under the curve (AUC) of 0.89, indicating excellent discriminative ability. The optimal cutoff was 1.387 ng/mL, which provided 85.7% sensitivity and 82.6% specificity (Fig. 2). Kaplan–Meier survival analysis using the cohort median (1.23 ng/mL) as the cutoff revealed significantly shorter metastasis-free survival in patients with higher gp100 levels ($P = 0.001$, log-rank test). This supports the potential prognostic value of serum gp100 at baseline (Supplementary Fig. S1).

Finally, in the multivariate logistic regression analysis including both molecular classification and serum gp100 levels, the latter remained a strong independent predictor of metastasis (odds ratio = 3849,931; 95% confidence interval [CI], 9.66–1,534,205.981; $P = 0.007$). The model explained 59% of the variance (Cox and Snell $R^2 = 0.588$), thereby

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reinforcing the potential utility of gp100 as a standalone biomarker for early risk stratification in UM (Supplementary Table S1).

Discussion

This study evaluated the prognostic value of serum gp100 in patients with UM without metastasis at the time of diagnosis. Higher gp100 concentrations were significantly associated with subsequent metastatic progression ($P < 0.001$), independently of clinical and genetic variables. Multivariate analysis confirmed gp100 as an independent prognostic factor ($P = 0.007$), and the ROC curve demonstrated strong discriminative power (AUC = 0.89). These findings support the use of gp100 as an accessible, noninvasive, and potentially valuable tool for early metastatic risk stratification in UM.

gp100 is a structural transmembrane glycoprotein that is essential for melanosome maturation, where it contributes to the formation of the fibrillar matrix that scaffolds melanin synthesis and deposition. It is one of the few known proteins capable of forming functional amyloid fibrils in vivo through the processing of its luminal domain, which enables ordered intramelanosomal assembly.[18] The expression of gp100 is restricted to the melanocytic lineage and remains elevated in various melanoma subtypes, including UM. At the molecular level, gp100 is regulated by the transcription factor MITF, which also controls genes such as *TYR* and *MART-1* that are involved in pigmentation, proliferation, and migration.[17] gp100 overexpression has been linked to an “activated melanocyte” phenotype, which is characterized by enhanced invasiveness, motility, and plasticity and is associated with a higher risk of tumor dissemination.[27] Furthermore, cytoplasmic processing of gp100 allows the generation of peptide fragments that are presented via major histocompatibility complex class I, which makes it an immunologically relevant tumor antigen. This property has been clinically leveraged through tebentafusp—a bispecific TCR–anti-CD3 fusion protein that targets gp100 presented by HLA-A*02:01. Tebentafusp is the first therapy to improve overall survival in metastatic UM, and this has validated gp100 as a stable, therapeutically actionable antigen.[20,28]

Proteomic analyses have identified gp100 as one of the most consistently secreted proteins in the UM secretome, suggesting that its presence in the serum may reflect subclinical tumor activity.[10] gp100 may be released through vesicular exocytosis or tumor cell necrosis and can be detected in biological fluids, such as blood and vitreous humor.[23] This biological characteristic may explain why patients without radiological evidence of metastasis but with more biologically active disease exhibit elevated serum gp100 levels. An important observation from this study is that gp100 concentration did not correlate with clinical (age, sex, tumor size) or genetic variables (monosomy 3, BAP1, 8q), reinforcing its role as an independent prognostic biomarker. Liquid biopsy strategies such as ctDNA, CTCs, and exosomal RNA have emerged as promising tools for prognostication in uveal melanoma. However, their sensitivity in early-stage disease remains limited. In a comprehensive review, Bustamante et al.[29] highlighted that ctDNA is often undetectable in nonmetastatic patients and rarely predicts progression in early clinical stages. Beasley et al.[30] similarly reported that ctDNA did not correlate with future metastasis and was only present in 68% of patients with radiologically confirmed disease. Zameer et al.[31] further emphasized the technical and biological limitations of ctDNA detection in uveal melanoma compared to cutaneous melanoma. In contrast, gp100 is a lineage-restricted glycoprotein that can be measured via a standardized ELISA assay and does not require tumor-informed mutation tracking. In our study, serum gp100 levels were significantly higher in patients who developed metastases despite the absence of clinical or genetic differences at baseline. These findings support the value of gp100 as an accessible, stable, and early biomarker of subclinical dissemination, particularly in centers where advanced molecular diagnostics are not available.

These findings are consistent with a previous study wherein gp100 was nearly undetectable in healthy individuals (mean, 0.00 ng/mL; 95% CI, 0.00–0.08 ng/mL) and in patients with choroidal nevi (mean, 0.31 ng/mL; 95% CI, 0.25–0.37 ng/mL) but was clearly elevated in patients with metastatic disease (mean, 2.20 ng/mL).[23] Unlike that study, which focused on advanced stages, the present work demonstrates that gp100 can predict metastasis even in early clinical stages and function as a marker of tumor behavior, regardless of genetic status. Clinically, gp100 measurement could be integrated into follow-up protocols to identify patients who are at higher risk of dissemination. ELISA-based quantification is simple, reproducible, and cost-effective and enables clinical adoption. Moreover, gp100 enables noninvasive longitudinal monitoring—which is particularly relevant in UM, where metastasis frequently develops silently and is diagnosed late. In the current landscape, following the approval of tebentafusp, which exploits gp100 expression for T-cell redirection, tools such as serum gp100 measurement may help identify patients with high antigenic load or guide early therapeutic strategies in those with increased gp100 levels.[21,32,33]

This study had some limitations. The moderate sample size and single-center design may limit the generalizability of the results. Although the follow-up duration (mean, 24.6 months) was clinically meaningful, the incidence of late-onset metastases may have been underestimated. Furthermore, in the absence of an internationally validated clinical cutoff for gp100, future studies are needed to establish a cutoff value before routine implementation. Future research should validate these findings in larger, multicenter cohorts to assess gp100 dynamics through longitudinal sampling after local treatment. The diagnostic integration of serum gp100 with other circulating biomarkers—such as ctDNA, microRNAs, or proteomic signatures—may improve risk stratification and facilitate the development of more personalized prognostic models in UM.

Beyond its structural and prognostic role, recent studies have shown that gp100 may actively modulate the immune microenvironment. Tebentafusp not only redirects T cells against gp100-positive tumor cells but also induces the infiltration of antigen-presenting cells and enhances tumor immunogenicity, even in tumors that are initially classified as “cold.”[34] This immunomodulatory effect may be particularly relevant for patients with elevated serum gp100 levels, as it could reflect an antigen-rich tumor profile that is more responsive to immunotherapy. Moreover, compared to other antigens such as MART-1 or PRAME, gp100 expression is notably stable across melanocytic tumors and is consistently retained even in dedifferentiated states. This has been confirmed both in modern molecular analyses and in earlier histopathologic studies of antigen heterogeneity in uveal melanoma.[35–37] This reinforces its reliability not only for immunotherapeutic targeting but also as a biomarker for dynamic disease monitoring. Unlike ctDNA, which may vary depending on tumor shedding and DNA integrity, gp100 may offer a steadier indicator of biological activity, tumor visibility, and immune engagement.

Given the liver's predominant role as the site of metastasis in UM—which has been reported in more than 90% of cases—circulating gp100 may serve as an early indicator of hepatic micrometastases, which are frequently undetectable on conventional imaging in the early dissemination phases.[5] The hepatic tropism of UM metastases has been linked to tumor–endothelial interactions, integrin expression, and immunosuppressive premetastatic niches, which further supports the role of soluble gp100 as a reflection of active systemic dissemination. Finally, proteomic studies suggest that gp100 is secreted alongside molecules that are involved in angiogenesis, matrix remodeling, and immune suppression, highlighting its potential role in early metastatic processes.[38]

In conclusion, longitudinal assessment of the serum gp100 level could help identify minimal residual disease, and its combination with transcriptomic or exosomal markers might further enhance the prediction of relapse or therapy failure.

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